# Chetan Mulge

**Software Development Engineer (Backend / Full Stack) | AI Engineer**

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## Professional Summary

Computer Science postgraduate (MCA, 2026) with verified industry internship experience in web development, backend systems, and AI automation workflows. Strong grounding in Data Structures and Algorithms, Object-Oriented Programming, and DBMS. IEEE-published researcher in Document AI and Computer Vision, having engineered custom YOLOv8 and EasyOCR pipelines achieving 85% detection accuracy and 0.00 Character Error Rate on Kannada script. Certified by OpenAI Academy in Agents and Workflows and OneRoadmap in AI Engineering; actively builds production automations using n8n, Webhooks, and Gemini APIs.

## Technical Skills

**Languages**: Java, Python, JavaScript (ES6+), SQL, C
**Backend & Web Development**: Node.js, Express.js, RESTful APIs, React, Next.js, HTML5, CSS3
**AI, ML & Automations**: n8n, Webhooks, Gemini API, OpenAI Agents & Workflows, YOLOv8, OpenCV, EasyOCR, PyTorch
**Databases & Tools**: MongoDB, MySQL, Google Sheets CRM, Git, GitHub, Linux, VS Code, Google Colab, XAMPP
**Core CS Fundamentals**: Data Structures & Algorithms (DSA), OOP, DBMS, Operating Systems, Computer Networks

## Experience

**CodTech IT Solutions** — Remote
*Web Development Intern* — *May 2025 – Jun 2025*

- Developed and styled interactive, responsive front-end web components adhering to modern UI standards.
- Implemented backend integration routines and handled client-server API interactions to ensure cross-device consistency.
- Maintained clean, reviewable Git version control practices, adhering strictly to enterprise task workflows (Verification ID: CT06DK547).

**iStudio** — Pune, India (Remote)
*Website Design and Development Intern* — *May 2025 – Jul 2025*

- Selected via the Internship Common Aptitude Test (iCAT) for an intensive 2-month professional web development role.
- Designed and implemented dynamic web layouts, improving UI/UX responsiveness and client-side data handling routines.
- Collaborated on full-stack application requirements, building modular and scalable code structures for live project delivery.

## Key Projects

**WhatsApp AI Lead Qualification** | *n8n, Gemini API, Webhooks, Google Sheets, Automation*

- Built an end-to-end automated lead-qualification workflow utilizing WhatsApp webhook triggers and n8n orchestration.
- Integrated Gemini API for natural language intent detection, user context parsing, and dynamic response generation.
- Connected Google Sheets as an automated, low-latency CRM store for capturing structured lead details and status updates.
- Validated bilingual Kannada–English conversation flows for a property-services client to ensure seamless lead nurturing.

**Multilingual Scene Text Detection & Recognition (IEEE Published)** | *Python, YOLOv8, OpenCV, EasyOCR*

- Engineered a bilingual Kannada–English scene text detection pipeline to localize and extract text from dynamic video footage.
- Curated and annotated a custom 500+ image dataset using LabelImg; automated frame extraction and preprocessing scripts.
- Trained a custom YOLOv8 model yielding 85% detection accuracy and 0.82 F1-score; achieved 0.00 Character Error Rate (CER) on Kannada text using EasyOCR; co-authored research paper in an IEEE conference.

**Skillverse – Learning Management System** | *React, Node.js, Express.js, MongoDB, REST APIs*

- Built backend services for a MERN LMS featuring secure JWT authentication, session handling, and Role-Based Access Control (RBAC).
- Designed and deployed RESTful APIs with a normalized MongoDB schema, enhancing query response efficiency for course browsing.

**Pharmacy Management System** | *Java, MySQL, JDBC, OOP*

- Developed a desktop application using Java OOP architecture and a normalized MySQL schema managing 1,000+ medicine records.
- Automated inventory tracking, low-stock threshold alerts, and instant billing generation, reducing manual administrative effort by 30%.

## Education

**Central University of Karnataka** — Kalaburagi, India
*Master of Computer Applications (MCA)* — *Aug 2024 – May 2026*

**SharnBasava University** — Kalaburagi, India
*Bachelor of Computer Applications (BCA)* – ***CGPA: 9.0/10.0 (84%)*** — *Aug 2021 – May 2024*

**Gurunagalingeshwar PU College** — Kalaburagi, India
*Pre-University Course (PUC – 12th)* – ***70%*** — *Jun 2019 – May 2021*

**Aradhana English Medium School** — Kalaburagi, India
*Secondary School Leaving Certificate (SSLC – 10th)* – ***70.08%*** — *Passed 2019*

## Certifications & Research

**IEEE Conference Publication**: Co-Author, Multilingual Scene Text Detection and Recognition Pipeline
**OpenAI Academy**: Agents and Workflows (ID: md7k90m7c4, Jul 2026) | Applied AI Foundations (ID: 0f6paimmd1, Jul 2026)
**OneRoadmap**: AI Engineer (ID: CERT-6CEAFA21, Jul 2026) | Data Analyst (ID: CERT-5F329963, Apr 2025)
**Technical Foundations**: Introduction to Java Programming (Coursera) | SQL for Beginners (Great Learning)